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# META-DIFFUSION: EVIDENCE-AUDITED PLAN-ALIGNED RESULTS AND FITZPATRICK17K SOURCE SPECIFICITY

**Anonymous authors**

Paper under double-blind review

## ABSTRACT

We report an evidence audit of a documented fixed experiment sequence designed to distinguish reusable population transport from genuinely source-task-specific transport. The sequence audits the data and frozen method (P0), raises Fitzpatrick17k task resolution from diseases to pathology families (P1), introduces hierarchical wrong-source controls (P2), tests a source-only task-separability explanation (P3), repeats the family protocol under a stronger Fitzpatrick I–III to V–VI shift (P4), audits the completed CIFAR package (P5), and applies a strict gate to downstream demographic augmentation (P6). On Fitzpatrick17k, correct-source transport improves energy MMD over a population-only control, but no correct-versus-wrong source difference is resolved under either phenotype definition. The frozen family classifier fails its documented per-family validity rule. Moreover, an exploratory real-image audit shows that the phenotype classifier has near-zero target-class recall, so neither semantic nor phenotype fidelity is validated. Source-only task separability does not provide a supported explanation. In contrast, audited CIFAR evidence retains a clear correct-versus-wrong-source advantage at  $K_T = 0$ , survives FiLM conditioning, and supports an empirical advantage for linear over nonlinear transport on the tested budgets. The resulting claim is therefore deliberately narrow: source-specific coordinate transport is supported in the controlled CIFAR setting, whereas Fitzpatrick17k provides only a population-level distributional diagnostic, not validated task-specific demographic transport.

## 1 SCOPE AND REPORTING CONVENTION

The core model, training recipe, linear transport, and no-refinement headline inference were documented before the saved family-level outcomes according to local file timestamps. This is not an external immutable pre-registration record. No new head, larger coordinate, nonlinear transport, or architecture rescue was introduced. All within-task comparisons use the same support draw, real target query, and DDIM noise. Repeats reduce Monte Carlo noise but are not treated as independent semantic tasks.

Raw-performance tables report absolute means  $\pm 95\%$  confidence-interval half-widths. A separate comparison table reports absolute paired differences. For lower-is-better distances, the comparison is

$$\Delta_d = d(\text{comparator}, \mathcal{D}_T) - d(\text{correct source+transport}, \mathcal{D}_T), \quad (1)$$

so positive values favour correct-source transport. For higher-is-better semantic or phenotype consistency, the subtraction is reversed, again making positive values favour correct transport. A superscript \* is attached directly to a paired result only when its 95% interval excludes zero. No star is placed on an unpaired raw mean.

The reported measurements are: phenotype target share and semantic consistency (higher is better), and fixed phototype-statistic distance, sliced Wasserstein distance (SW), and energy MMD (lower is better). A frozen predictor is a measurement instrument, not ground truth. Failure of its independent validity gate prevents a semantic claim and is not replaced post hoc by a distributional metric. A

dagger (†) marks the saved phenotype target-share output, which is retained for traceability but is not a validated fidelity measure after the evaluator audit in Section 7.

## 2 P0: PREREQUISITE AND REPRODUCIBILITY GATE

The experiment used the verified SFU Fitzpatrick17k-Renamed archive as a byte source. Images were matched to the official CSV using the supplied MD5 hashes and filtered with the Fitzpatrick17k-C allow-list. Known near-duplicate components were collapsed before task allocation. The official schema contains no subject identifier; consequently patient-level leakage cannot be checked directly, and hash/near-duplicate filtering is the strongest available safeguard.

Table 1: P0 dataset, environment, and frozen-protocol audit.

| Audit item                                      | Verified value              |
|-------------------------------------------------|-----------------------------|
| Official CSV / archive rows                     | 16,577 / 16,577             |
| Byte-verified rows retained by Fitzpatrick17k-C | 11,394                      |
| Retained rows with Fitzpatrick I–VI             | 11,098                      |
| P1 eligible pathology families (target IV–VI)   | 6                           |
| P4 eligible pathology families (target V–VI)    | 5                           |
| P1 task-level folds / P4 cyclic folds           | 3 / 5                       |
| Training / evaluation seeds                     | 12,345 / 4,321              |
| Python / PyTorch / CUDA                         | 3.12.13 / 2.11.0 / 12.8     |
| GPU                                             | NVIDIA RTX 5080, 16,303 MiB |
| Frozen coordinate dimension / source budget     | $k = 16 / M_S = 32$         |
| Protocol / first P1 output (local time)         | 16:41 / 17:16, 21 Sep 2026  |

The official CSV SHA-256 is 851DFAAB...D4A1CD; the verification-manifest SHA-256 is 4AA4FC1C...58DF3. All manifests passed on-disk, population-side, taxonomy, duplicate, and capacity checks. Target support is disjoint from target query and nested across  $K_T \in \{1, 2, 5, 10, 20\}$ . The family-level semantic-instrument check remains an outcome-validity gate and was therefore not used to select tasks or folds.

Three families were excluded from P1 by the fixed count rule: malignant dermal (53 source, 43 target), benign melanocyte (118/38), and malignant cutaneous lymphoma (99/38). P4 also excludes malignant melanoma because only 38 V–VI targets remain. These exclusions were determined from counts, not model outcomes.

**P0 conclusion.** The prerequisite gate passed with two limits: patient identity is unavailable, and the protocol chronology is supported by local timestamps rather than external registration. Data provenance, exclusions, task folds, checkpoints, seeds, comparison direction, and independent statistical units are traceable in the audit appendix.

## 3 P1: PATHOLOGY-FAMILY SEMANTIC RESOLUTION

The task is the official `nine_partition_label`; source is Fitzpatrick I–III and target is IV–VI. Six families meet the pre-model requirement of at least 64 clean source and 45 clean target images. Three task-level folds test every family exactly once. Model selection uses only the validation family in each fold and chooses training steps 16,000, 10,000, and 10,000. Evaluation uses 96 generated images per cell, DDIM-50, eight paired repeats, and no target-time refinement.

### 3.1 ABSOLUTE ZERO-SHOT PERFORMANCE

The semantic evaluator obtains overall fold accuracies of 67.6–68.7%, but its minimum outer-test-family accuracies are 0.0%, 0.0%, and 15.3%, below the locked 25% per-family threshold. Hence the primary correct-minus-wrong semantic effect of exactly zero is uninformative rather than evidence of equivalence. The distributional correct-versus-wrong effects all include zero. The phenotype-share effect also includes zero, but that instrument is separately judged unreliable in Section 7.

Table 2: P1 raw performance at  $K_T = 0$  over six outer-test pathology families. Entries are absolute family means  $\pm$  95% CI. Phenotype share and family consistency are higher-is-better; all three distances are lower-is-better.

| Method                            | Phenotype share $\uparrow$ | Family consistency $\uparrow$ | Statistic dist. $\downarrow$ | SW $\downarrow$     | Energy MMD $\downarrow$ |
|-----------------------------------|----------------------------|-------------------------------|------------------------------|---------------------|-------------------------|
| Source reuse                      | 0.0000 $\pm$ 0.0000        | 0.1651 $\pm$ 0.3227           | 1.7452 $\pm$ 0.2089          | 0.3081 $\pm$ 0.0192 | 5.7946 $\pm$ 1.0078     |
| Population only                   | 0.0000 $\pm$ 0.0000        | 0.1641 $\pm$ 0.3216           | 3.0504 $\pm$ 0.2209          | 0.2683 $\pm$ 0.0110 | 6.8472 $\pm$ 1.0089     |
| Wrong-family source + relation    | 0.0000 $\pm$ 0.0000        | 0.1656 $\pm$ 0.3230           | 1.7185 $\pm$ 0.2170          | 0.3064 $\pm$ 0.0200 | 5.7468 $\pm$ 1.0114     |
| Correct-family source + transport | 0.0002 $\pm$ 0.0004        | 0.1656 $\pm$ 0.3230           | 1.7188 $\pm$ 0.2178          | 0.3063 $\pm$ 0.0199 | 5.7436 $\pm$ 1.0026     |

Table 3: P1 paired effects at  $K_T = 0$ . Every entry is oriented so positive values favour correct-family transport. A star means the paired 95% CI excludes zero.

| Comparator ( $\Delta > 0$ : correct-family transport) | Phenotype share        | Family consistency     | Statistic dist.         | SW                      | Energy MMD              |
|-------------------------------------------------------|------------------------|------------------------|-------------------------|-------------------------|-------------------------|
| Source reuse                                          | +0.00022 $\pm$ 0.00043 | +0.00043 $\pm$ 0.00054 | +0.02635 $\pm$ 0.01482* | +0.00178 $\pm$ 0.00218  | +0.05099 $\pm$ 0.05141  |
| Population only                                       | +0.00022 $\pm$ 0.00043 | +0.00152 $\pm$ 0.00191 | +1.33162 $\pm$ 0.08916* | -0.03795 $\pm$ 0.02043* | +1.10360 $\pm$ 0.49696* |
| Wrong-family source + relation                        | +0.00022 $\pm$ 0.00043 | +0.00000 $\pm$ 0.00000 | -0.00029 $\pm$ 0.00160  | +0.00007 $\pm$ 0.00091  | +0.00317 $\pm$ 0.01925  |

### 3.2 TARGET-ONLY DATA-EFFICIENCY COMPARISON

Because the headline correct, wrong, source-reuse, and population-only coordinates receive no target-time refinement, they are invariant to  $K_T$ . Nonzero target budgets therefore test the target-only comparator rather than further adapting transport. Code audit confirms that target-only does consume the nested  $K_T$  support: it encodes `batch.tgt_support` and then refines on those same images. Target-only is N/A at  $K_T = 0$ .

Table 4: P1 target-only raw distances and paired effects against the fixed correct-family transport. Within each metric, the first row is target-only raw performance and the second is  $\Delta = d(\text{target only}) - d(\text{correct transport})$ ; positive supports correct transport.

| Metric          | Quantity                      | $K_T = 1$            | $K_T = 2$            | $K_T = 5$             | $K_T = 10$            | $K_T = 20$            |
|-----------------|-------------------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
| Statistic dist. | Target only raw               | 1.7322 $\pm$ 0.2193  | 1.7306 $\pm$ 0.2201  | 1.7412 $\pm$ 0.2117   | 1.7369 $\pm$ 0.2147   | 1.7356 $\pm$ 0.2158   |
|                 | Paired $\Delta$ (+ = correct) | +0.0134 $\pm$ 0.0227 | +0.0118 $\pm$ 0.0127 | +0.0224 $\pm$ 0.0119* | +0.0180 $\pm$ 0.0080* | +0.0168 $\pm$ 0.0092* |
| SW              | Target only raw               | 0.3045 $\pm$ 0.0194  | 0.3055 $\pm$ 0.0191  | 0.3041 $\pm$ 0.0190   | 0.3047 $\pm$ 0.0204   | 0.3039 $\pm$ 0.0194   |
|                 | Paired $\Delta$ (+ = correct) | -0.0018 $\pm$ 0.0037 | -0.0007 $\pm$ 0.0012 | -0.0022 $\pm$ 0.0018* | -0.0016 $\pm$ 0.0014* | -0.0024 $\pm$ 0.0020* |
| Energy MMD      | Target only raw               | 5.7208 $\pm$ 0.9739  | 5.7303 $\pm$ 0.9850  | 5.6749 $\pm$ 0.9871   | 5.7091 $\pm$ 1.0350   | 5.6900 $\pm$ 1.0112   |
|                 | Paired $\Delta$ (+ = correct) | -0.0228 $\pm$ 0.0745 | -0.0133 $\pm$ 0.0315 | -0.0687 $\pm$ 0.0379* | -0.0345 $\pm$ 0.0374  | -0.0536 $\pm$ 0.0371* |

The distance metrics disagree in direction, and target-only itself changes little with  $K_T$ . Consequently, no scientifically meaningful “transport at  $K_T = 2$  equals target-only at  $K_T = 10$ ” statement is supported.

**P1 conclusion.** Raising semantic resolution to pathology families does not recover a resolved source-specific advantage. The primary result is inconclusive because its evaluator fails the validity gate; distributional diagnostics show no resolved correct-versus-wrong effect, and phenotype fidelity is not validated.

## 4 P2: HIERARCHICAL WRONG-SOURCE CONTROL

The frozen disease-level checkpoint is evaluated on four test diseases with correct disease, wrong disease in the same pathology family, wrong disease in a different family, source reuse, and population-only conditions. Wrong sources are deterministic and selected without using model outcomes.

Table 5: P2 raw zero-shot performance over four disease tasks. Consistency measures are higher-is-better; distances are lower-is-better.

| Method                          | Phenotype share <sup>†</sup> ↑ | Disease consistency ↑ | Statistic dist. ↓ | SW ↓            | Energy MMD ↓    |
|---------------------------------|--------------------------------|-----------------------|-------------------|-----------------|-----------------|
| Source reuse                    | 0.1390 ± 0.0103                | 0.0199 ± 0.0224       | 2.5248 ± 0.4420   | 0.3642 ± 0.0448 | 7.1765 ± 1.8278 |
| Population only                 | 0.0000 ± 0.0000                | 0.0000 ± 0.0000       | 4.0465 ± 0.3111   | 0.2867 ± 0.0094 | 8.4246 ± 1.4825 |
| Wrong disease, same family      | 0.1423 ± 0.0101                | 0.0199 ± 0.0224       | 2.5020 ± 0.4478   | 0.3638 ± 0.0454 | 7.1466 ± 1.8509 |
| Wrong disease, different family | 0.1426 ± 0.0098                | 0.0205 ± 0.0231       | 2.5025 ± 0.4462   | 0.3639 ± 0.0454 | 7.1444 ± 1.8488 |
| Correct disease + transport     | 0.1419 ± 0.0094                | 0.0199 ± 0.0221       | 2.5030 ± 0.4480   | 0.3638 ± 0.0456 | 7.1429 ± 1.8566 |

Table 6: P2 hierarchical paired effects at  $K_T = 0$ . Positive values favour the first named condition. A star denotes a paired 95% interval excluding zero.

| Comparison ( $\Delta > 0$ : first condition) | Phenotype share     | Disease consistency | Statistic dist.     | SW                  | Energy MMD          |
|----------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Correct vs. same-family wrong                | -0.00033 ± 0.00283  | +0.00000 ± 0.00104  | -0.00101 ± 0.00777  | +0.00009 ± 0.00062  | +0.00365 ± 0.01937  |
| Same-family vs. different-family wrong       | -0.00033 ± 0.00191  | -0.00065 ± 0.00074  | +0.00053 ± 0.00532  | +0.00004 ± 0.00038  | -0.00219 ± 0.01319  |
| Correct vs. population only                  | +0.14193 ± 0.00938* | +0.01986 ± 0.02214  | +1.54346 ± 0.14526* | -0.07706 ± 0.03919* | +1.28164 ± 0.52335* |

The disease instrument reaches 20.8% overall but only 53.1%, 0%, 6.2%, and 0% on the four individual test diseases. It is therefore unsuitable for a strong semantic ordering. The correct/same-family/different-family distance contrasts are also numerically small and all include zero. Population-only again produces low pixel SW while failing phenotype and energy MMD checks, demonstrating why no single pixel metric is sufficient.

Table 7: P2 target-only raw distances and reporting-only paired effects versus fixed correct transport over four tasks.  $\Delta = d(\text{target only}) - d(\text{correct})$ ; positive supports correct. The  $K_T = 0$  target-only entry is N/A.

| Quantity     | $K_T = 1$        | $K_T = 2$        | $K_T = 5$        | $K_T = 10$        | $K_T = 20$       |
|--------------|------------------|------------------|------------------|-------------------|------------------|
| SW raw       | 0.3617 ± 0.0439  | 0.3634 ± 0.0443  | 0.3622 ± 0.0442  | 0.3626 ± 0.0455   | 0.3625 ± 0.0442  |
| SW $\Delta$  | -0.0020 ± 0.0021 | -0.0003 ± 0.0015 | -0.0015 ± 0.0022 | -0.0012 ± 0.0005* | -0.0012 ± 0.0017 |
| MMD raw      | 7.1134 ± 1.8041  | 7.1633 ± 1.8157  | 7.1195 ± 1.8139  | 7.1320 ± 1.8454   | 7.1369 ± 1.8166  |
| MMD $\Delta$ | -0.0295 ± 0.0687 | +0.0204 ± 0.0479 | -0.0234 ± 0.0706 | -0.0109 ± 0.0218  | -0.0061 ± 0.0492 |

**P2 conclusion.** Neither correct > same-family wrong > different-family wrong nor a family-only ordering is resolved. The null hierarchy is not an equivalence result.

## 5 P3: OUTCOME-INDEPENDENT SOURCE-TASK SEPARABILITY

Official ImageNet ResNet-18 penultimate features were extracted from source-population images only, L2-normalised per image, and averaged by disease. Separation is Euclidean distance to the pre-assigned different-family wrong source from P2.

Table 8: P3 source-only separation and held-out semantic effect.

| Target disease       | Separation | Correct-minus-wrong effect |
|----------------------|------------|----------------------------|
| Basal cell carcinoma | 0.2488     | 0.0000                     |
| Neurofibromatosis    | 0.2807     | 0.0000                     |
| Pityriasis rosea     | 0.3289     | -0.0026                    |
| Vitiligo             | 0.2971     | 0.0000                     |

Spearman  $\rho = -0.775$  with a task-bootstrap interval  $[-1.000, -0.775]$ . Only 6,793 of 10,000 bootstrap draws have defined rank variance because  $n = 4$  and three outcome effects are exactly zero. Moreover, the outcome is measured by the weak disease instrument described above.

**P3 conclusion.** Greater source-task separability does not predict a larger correct-source advantage in the locked analysis. The result is unsupported/unresolved, not a negative causal mechanism, and task identifiability should not be the paper’s central explanation.

## 6 P4: STRONGER PHENOTYPE SEPARATION

P4 was triggered because P1 was inconclusive and the documented pre-outcome counts were adequate. The only domain change is target Fitzpatrick V–VI; source remains I–III. Five families are tested in five cyclic outer folds, with each family’s two appearances averaged before inference. Validation-only selection chooses steps 16,000, 4,000, 4,000, 4,000, and 10,000. The architecture is a  $32 \times 32$  additive-basis small U-Net (0.72M parameters), set encoder (0.33M), linear transport (0.3K), coordinate dimension 16, and no target-time refinement.

Table 9: P4 raw performance at  $K_T = 0$  over five pathology families. The stronger target is Fitzpatrick V–VI.

| Method                       | Phenotype share <sup>†</sup> ↑ | Family consistency ↑ | Statistic dist. ↓ | SW ↓            | Energy MMD ↓    |
|------------------------------|--------------------------------|----------------------|-------------------|-----------------|-----------------|
| Source reuse                 | 0.0311 ± 0.0359                | 0.2033 ± 0.3807      | 1.8099 ± 0.2613   | 0.2649 ± 0.0446 | 4.5031 ± 0.7991 |
| Population only              | 0.0000 ± 0.0000                | 0.1997 ± 0.3909      | 2.2577 ± 0.4548   | 0.2527 ± 0.0275 | 5.4122 ± 1.1998 |
| Wrong-task source + relation | 0.0320 ± 0.0369                | 0.2033 ± 0.3807      | 1.7943 ± 0.2765   | 0.2637 ± 0.0454 | 4.4233 ± 0.8218 |
| Correct source + transport   | 0.0315 ± 0.0363                | 0.2031 ± 0.3808      | 1.7950 ± 0.2769   | 0.2638 ± 0.0461 | 4.4244 ± 0.8362 |

Table 10: P4 paired effects at  $K_T = 0$ . All entries are oriented so positive values favour correct-source transport.

| Comparator ( $\Delta > 0$ : correct transport) | Phenotype share    | Family consistency | Statistic dist.    | SW                 | Energy MMD          |
|------------------------------------------------|--------------------|--------------------|--------------------|--------------------|---------------------|
| Source reuse                                   | +0.00039 ± 0.00051 | −0.00013 ± 0.00102 | +0.01494 ± 0.01616 | +0.00117 ± 0.00272 | +0.07874 ± 0.08272  |
| Population only                                | +0.03151 ± 0.03635 | +0.00339 ± 0.01614 | +0.46273 ± 0.46930 | −0.01112 ± 0.02383 | +0.98779 ± 0.70849* |
| Wrong-task source + relation                   | −0.00052 ± 0.00074 | −0.00013 ± 0.00026 | −0.00068 ± 0.00273 | −0.00003 ± 0.00134 | −0.00107 ± 0.03907  |

The phenotype instrument’s saved aggregate outer-test accuracy is 81.2–92.7%, but this number is not class-balanced. The exploratory audit in Section 7 finds only 0–6.3% real-target recall. It therefore cannot validate phenotype fidelity. The semantic instrument has 69.9–70.5% overall accuracy, yet worst-family accuracies of 0.0%, 0.0%, 0.0%, 12.7%, and 2.5%, so every fold fails the locked per-family rule. Pooled KID is consequently skipped. Correct transport has a paired energy-MMD advantage over population-only, but the correct-versus-wrong-task effect is  $-0.0011 \pm 0.0391$  and is unresolved. The same near-zero pattern holds for the other distributional metrics.

Table 11: P4 target-only raw distances and paired effects against fixed correct transport. For each metric,  $\Delta = d(\text{target only}) - d(\text{correct transport})$ ; positive values support correct transport.

| Metric          | Quantity                      | $K_T = 1$         | $K_T = 2$        | $K_T = 5$         | $K_T = 10$       | $K_T = 20$       |
|-----------------|-------------------------------|-------------------|------------------|-------------------|------------------|------------------|
| Statistic dist. | Target only raw               | 1.8082 ± 0.2694   | 1.8050 ± 0.2706  | 1.8044 ± 0.2691   | 1.7997 ± 0.2702  | 1.7972 ± 0.2689  |
|                 | Paired $\Delta$ (+ = correct) | +0.0132 ± 0.0109* | +0.0100 ± 0.0121 | +0.0094 ± 0.0086* | +0.0047 ± 0.0071 | +0.0022 ± 0.0106 |
| SW              | Target only raw               | 0.2597 ± 0.0413   | 0.2622 ± 0.0440  | 0.2624 ± 0.0443   | 0.2632 ± 0.0449  | 0.2630 ± 0.0447  |
|                 | Paired $\Delta$ (+ = correct) | −0.0041 ± 0.0052  | −0.0016 ± 0.0022 | −0.0013 ± 0.0020  | −0.0006 ± 0.0014 | −0.0008 ± 0.0017 |
| Energy MMD      | Target only raw               | 4.3571 ± 0.7399   | 4.3970 ± 0.7715  | 4.4157 ± 0.7973   | 4.4171 ± 0.7974  | 4.3974 ± 0.7895  |
|                 | Paired $\Delta$ (+ = correct) | −0.0673 ± 0.1207  | −0.0274 ± 0.0679 | −0.0086 ± 0.0460  | −0.0073 ± 0.0406 | −0.0270 ± 0.0576 |

**P4 conclusion.** The V–VI cohort does not yield a resolved correct-versus-wrong source effect. Correct transport improves energy MMD relative to population-only, but wrong-source transport gives the same distributional reading within uncertainty. Because P4 has five eligible families where P1 has six, this between-study pattern cannot be attributed exclusively to phenotype separation. Under the documented stopping rule, demographic model tuning ends here.

## 7 PHENOTYPE-MEASUREMENT AUDIT

Phenotype target share is a hard class-1 fraction from a binary TinyCNN: class 0 is I–III; class 1 is IV–VI (P1) or V–VI (P4). It has no abstention or calibration. Real and generated images use the same RGB  $[-1, 1]$  preprocessing. The saved evaluator reports only overall accuracy on the naturally imbalanced test-family images, and its construction did not explicitly seed PyTorch before weight initialization.

We therefore ran a post-outcome, *exploratory* sanity check using the same training recipe, frozen manifests, and an explicit torch seed. It does not replace any confirmatory cell. On real outer-test

images, P1 target recall is 0% in all three folds; P4 target recall is 0.0%, 0.5%, 0.0%, 6.3%, and 0.9%, while source recall is 99.8–100%.

Table 12: Exploratory real-image phenotype-instrument audit. Aggregate accuracy is high because source images dominate; balanced accuracy reveals near-constant source prediction.

| Protocol          | Aggregate accuracy range | Target recall range | Balanced accuracy range |
|-------------------|--------------------------|---------------------|-------------------------|
| P1, I–III → IV–VI | 72.7–81.1%               | 0.0%                | 50.0%                   |
| P4, I–III → V–VI  | 81.3–92.7%               | 0.0–6.3%            | 50.0–53.1%              |

**Measurement conclusion.** The saved near-zero generated target shares cannot validate either phenotype failure or success, because the classifier itself has essentially no real-target sensitivity in this check. Fixed phototype statistics, SW, and MMD remain distributional diagnostics; none is treated as proof of demographic identity.

## 8 P5: FROZEN CIFAR PACKAGE

P5 was audited from saved matched per-episode artifacts. Tables report both raw performance and paired effects; every effect is oriented so positive supports the named preferred method.

Table 13: CIFAR zero-shot source controls ( $n = 60$  episodes). For SW, effect is comparator minus correct; for class consistency, effect is correct minus comparator. Positive supports correct transport; stars mark paired 95% intervals excluding zero.

| Method                     | SW raw ↓          | SW effect              | Fine-class raw ↑  | Class effect           |
|----------------------------|-------------------|------------------------|-------------------|------------------------|
| Source reuse               | $0.248 \pm 0.018$ | $+0.0384 \pm 0.0113^*$ | $0.113 \pm 0.023$ | $-0.0017 \pm 0.0090$   |
| Relation only              | $0.236 \pm 0.022$ | $+0.0260 \pm 0.0090^*$ | $0.051 \pm 0.008$ | $+0.0606 \pm 0.0210^*$ |
| Wrong source + relation    | $0.248 \pm 0.021$ | $+0.0386 \pm 0.0141^*$ | $0.052 \pm 0.010$ | $+0.0597 \pm 0.0204^*$ |
| Correct source + transport | $0.210 \pm 0.020$ | –                      | $0.111 \pm 0.023$ | –                      |

Table 14: FiLM source-dependence on SW. Each cell is comparator raw / paired  $\Delta$ ;  $\Delta > 0$  supports correct FiLM transport.

| Method            | $K_T = 1$ raw / $\Delta$                 | $K_T = 5$ raw / $\Delta$                 | $K_T = 20$ raw / $\Delta$                |
|-------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| Mean source       | $0.246 \pm 0.024 / +0.0397 \pm 0.0084^*$ | $0.248 \pm 0.020 / +0.0430 \pm 0.0091^*$ | $0.250 \pm 0.023 / +0.0430 \pm 0.0096^*$ |
| Shuffled source   | $0.257 \pm 0.025 / +0.0501 \pm 0.0154^*$ | $0.259 \pm 0.023 / +0.0533 \pm 0.0169^*$ | $0.261 \pm 0.024 / +0.0537 \pm 0.0167^*$ |
| Relation only     | $0.241 \pm 0.023 / +0.0342 \pm 0.0077^*$ | $0.242 \pm 0.020 / +0.0361 \pm 0.0082^*$ | $0.245 \pm 0.023 / +0.0380 \pm 0.0090^*$ |
| Correct transport | $0.207 \pm 0.022 / -$                    | $0.205 \pm 0.019 / -$                    | $0.207 \pm 0.021 / -$                    |

Fine-class consistency independently favors correct FiLM transport over all three controls at every budget (paired gains 0.0696–0.0773; every interval excludes zero). Thus the source dependence is not specific to additive-basis conditioning.

Table 15: Linear versus nonlinear FiLM transport.  $\Delta =$  nonlinear minus linear, so positive supports linear transport.

| $K_T$ | Nonlinear SW raw    | Linear SW raw       | Paired $\Delta$        |
|-------|---------------------|---------------------|------------------------|
| 1     | $0.2066 \pm 0.0224$ | $0.1973 \pm 0.0206$ | $+0.0093 \pm 0.0037^*$ |
| 5     | $0.2054 \pm 0.0185$ | $0.1942 \pm 0.0164$ | $+0.0112 \pm 0.0040^*$ |
| 20    | $0.2074 \pm 0.0214$ | $0.1987 \pm 0.0193$ | $+0.0087 \pm 0.0042^*$ |

Energy MMD also favors linear transport at all three budgets. This is evidence for the tested FiLM checkpoints and budgets, not a proof that global task-coordinate geometry is linear and not a non-inferiority claim.

Table 16: Full fine-tuning reference on SW ( $n = 180$  paired episodes). Effects are comparator minus transport, so positive supports transport and negative supports the comparator.

| $K_T$ | Target-only raw     | Transport raw       | Full-FT raw         | Target-only effect     | Full-FT effect         |
|-------|---------------------|---------------------|---------------------|------------------------|------------------------|
| 1     | $0.2027 \pm 0.0139$ | $0.1839 \pm 0.0122$ | $0.2675 \pm 0.0193$ | $+0.0189 \pm 0.0092^*$ | $+0.0836 \pm 0.0180^*$ |
| 5     | $0.1950 \pm 0.0148$ | $0.1896 \pm 0.0120$ | $0.2124 \pm 0.0183$ | $+0.0054 \pm 0.0069$   | $+0.0228 \pm 0.0202^*$ |
| 20    | $0.1840 \pm 0.0111$ | $0.1896 \pm 0.0122$ | $0.1465 \pm 0.0091$ | $-0.0056 \pm 0.0067$   | $-0.0431 \pm 0.0149^*$ |

Table 17: CIFAR task-scarcity SW.  $\Delta = \text{FiLM}$  minus basis, so positive supports basis and negative supports FiLM. Every arm has one training seed.

| Training tasks | Quantity         | $K_T = 1$              | $K_T = 5$              | $K_T = 20$             |
|----------------|------------------|------------------------|------------------------|------------------------|
| 50%            | Basis / FiLM raw | 0.2097/0.1912          | 0.2060/0.1851          | 0.2141/0.1920          |
|                | Paired $\Delta$  | $-0.0185 \pm 0.0108^*$ | $-0.0208 \pm 0.0114^*$ | $-0.0221 \pm 0.0122^*$ |
| 25%            | Basis / FiLM raw | 0.2005/0.2187          | 0.1957/0.2160          | 0.2017/0.2215          |
|                | Paired $\Delta$  | $+0.0181 \pm 0.0095^*$ | $+0.0203 \pm 0.0093^*$ | $+0.0198 \pm 0.0091^*$ |

The direction reverses between the one-seed 50% and 25% runs. At 25%, energy MMD also supports basis by 0.4668–0.4975 with intervals excluding zero, while fine-class consistency resolves no basis advantage. These are conditional observations, not general evidence of architecture superiority.

The optional disease-level Fitzpatrick representation check uses four test diseases, not the P1/P4 family cohorts. At 50% of its meta-training diseases, FiLM has lower fixed-statistic, SW, and energy-MMD distances than basis at every budget. For example, at  $K_T = 1$ , basis/FiLM SW is 0.351/0.330 and the FiLM advantage is  $0.0214 \pm 0.0048^*$ ; at  $K_T = 20$  it is  $0.0199 \pm 0.0057^*$ . The one-seed coordinate-compression diagnostic instead finds the basis  $k = 4$  lower than  $k = 16$  on SW by  $0.0030 \pm 0.0027^*$ ,  $0.0053 \pm 0.0031^*$ , and  $0.0038 \pm 0.0017^*$  at  $K_T = 1, 5, 20$ . These distributional results do not repair the invalid phenotype/semantic instruments.

For the full-task matched comparison, transported SW does not resolve a basis–FiLM difference at any tested budget. The basis uses 55,344 conditioning parameters versus 118,784 for FiLM (46.6%). Matched latency, FLOPs, and peak memory are **not measured** and are not inferred from parameter count.

**P5 conclusion.** The controlled CIFAR evidence supports source-specific zero-shot transfer and FiLM generality. Linear transport has lower SW/MMD than the tested nonlinear FiLM map on the measured budgets. Transport is favored over full fine-tuning at  $K_T = 1, 5$ , while full fine-tuning is favored at  $K_T = 20$ . The basis result is parameter-efficiency plus a single-seed scarcity observation, not established general generation-quality superiority.

## 9 P6: OPTIONAL DOWNSTREAM DEMOGRAPHIC AUGMENTATION

P6 was **not run—gate not met**. Its strict trigger required a clearly positive P1 correct-versus-wrong family-semantic result. P1 instead failed the semantic-instrument validity rule and showed no resolved correct-versus-wrong advantage on the valid secondary metrics. Launching or tuning a downstream classifier would therefore violate the plan.

## 10 DISCUSSION

The Fitzpatrick pattern is consistent across four complementary analyses. First, family resolution yields no resolved correct-versus-wrong source difference. Second, wrong diseases within and outside a family are not resolved as different. Third, a frozen source-only separability score does not provide a supported explanation. Fourth, the differently composed V–VI cohort also yields no resolved task-source specificity. A population-only energy-MMD comparison can show that a relation changes the generated distribution; it cannot show that the change preserves the requested family or reaches the target phenotype.

This does not imply that demographic transport is impossible, nor does it prove that all source tasks are equivalent. The semantic evaluator fails on several held-out families and diseases; the phenotype evaluator lacks target sensitivity; independent task counts are small; and patient identifiers are unavailable. The frozen method and available measurements therefore do not justify a real-data source-specific or phenotype-fidelity claim. In particular, an energy-MMD improvement over population-only is insufficient when wrong-task transport has the same reading and the identity instruments are invalid.

## 11 CONCLUSION

The executable plan is complete through its documented stopping conditions. On CIFAR, META-DIFFUSION supports the sequence

source identity matters  $\rightarrow$  linear transport is sufficient on tested budgets  
 $\rightarrow$  benefit is largest under target scarcity  $\rightarrow$  the principle survives FiLM.

On Fitzpatrick17k, the corresponding source-specific demographic extension is not supported. Correct and wrong task sources remain unresolved at disease and family resolution, including in the V–VI cohort. The energy-MMD comparison supports a distributional change relative to population-only, but neither the target phenotype nor task semantics is validated. The headline family result therefore remains inconclusive, not equivalent. No demographic architecture rescue or downstream augmentation is warranted under the documented protocol.

## A FROZEN CONFIGURATION AND ARTIFACT MAP

Table 18: Frozen family-level training and evaluation configuration.

| Component                           | Value                                      |
|-------------------------------------|--------------------------------------------|
| Image resolution / backbone width   | $32 \times 32 / 32$                        |
| Conditioning / coordinate dimension | additive basis / 16                        |
| Transport                           | linear, approximately 0.3K parameters      |
| Training                            | 20,000 steps, AdamW, lr $2 \times 10^{-4}$ |
| EMA / training seed                 | 0.999 / 12,345                             |
| Source support                      | $M_S = 32$                                 |
| Target budgets                      | $K_T = 0, 1, 2, 5, 10, 20$                 |
| Generation                          | DDIM-50, $\eta = 0$ , 96 images/cell       |
| Paired repeats / evaluation seed    | 8 / 4,321                                  |
| Headline target-time refinement     | disabled                                   |

## B PER-FAMILY SOURCE-SPECIFICITY EFFECTS

Table 19: Correct-versus-wrong source effects at  $K_T = 0$  for each outer-test family. Semantic  $\Delta$  is correct minus wrong; MMD  $\Delta$  is wrong minus correct. Positive supports correct transport. These task rows are the independent units behind the aggregate intervals; no per-row significance test is performed.

| Family              | P1 semantic $\Delta$ | P1 MMD $\Delta$ | P4 semantic $\Delta$ | P4 MMD $\Delta$ |
|---------------------|----------------------|-----------------|----------------------|-----------------|
| Benign dermal       | 0.0000               | -0.0303         | -0.0007              | -0.0737         |
| Benign epidermal    | 0.0000               | +0.0436         | 0.0000               | +0.0464         |
| Genodermatoses      | 0.0000               | +0.0004         | 0.0000               | +0.0183         |
| Inflammatory        | 0.0000               | +0.0106         | 0.0000               | -0.0016         |
| Malignant epidermal | 0.0000               | -0.0049         | 0.0000               | +0.0052         |
| Malignant melanoma  | 0.0000               | -0.0005         | –                    | –               |



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## C EVIDENCE AND REPRODUCIBILITY MAP

The report baseline is Git commit 2cd24b1aa79f8425c9d361c65d838751862a26e9. The P1-P4 aggregate JSON SHA-256 prefixes are respectively 4B55EAF5, 86622B4F, 28D54502, and BBC50FDD. Checkpoint choices are validation-only and recorded in runs/\*\_selection.json; P1 selects steps 16k/10k/10k and P4 selects 16k/4k/4k/4k/10k. Commands and exclusions are recorded in runs/execution\_log.md and prerequisite\_audit.md.

Machine-readable raw cells, family/disease-level values, checkpoint-selection records, manifests, hashes, commands, and complete tables are stored under experiments/meta\_diffusion\_final/. The principal files are protocol\_lock.md, prerequisite\_audit.md, metrics/p1\_family\_aggregate.json, metrics/p2\_hierarchical.json, metrics/p3\_task\_separability.json, metrics/p4\_v\_vi\_aggregate.json, and runs/execution\_log.md. Debug runs are explicitly named and excluded from every reported estimate. This revision adds revision\_audit/evidence\_manifest.md, phenotype\_metric\_audit.md, statistics\_audit.md, and p5\_claim\_evidence.md; the original report snapshot and generated diff are retained beside them.